

9.7 Recursion

sum of n numbers

~~if ($n == 1$)~~

~~return 1~~

else

return ~~1~~

~~sum = 0~~

if ($n == 1$)

return 1

else

return $n + \text{sum}(n-1)$

Q) Find out the applications of recursion. Is recursion still used in the present programming era? What are the languages that supports recursive calls.

A. Applications of Recursion

1. Tree traversal (Binary Trees, BSTs, Expression Trees)
2. Graph algorithms such as Depth-First Search (DFS)
3. Divide and conquer algorithms like Merge, Sort and Quicksort.
4. Backtracking problems such as Sudoku solver, N-Queens and Maze solving
5. Mathematical calculations like factorial, fibonacci, GCD and power functions
6. File and folder traversal in operating system
7. Parsing and compiler design for processing nested expressions.

8. Dynamic program (using recursion with memoization)

Yes, Recursion is still important because:

- It makes many algorithms simpler and easier to understand.
- It is commonly used in competitive programming, AI, compilers and data structures.
- Some languages optimize recursive calls (tail recursion), making recursion efficient in certain cases.
- However, for very deep recursion, iteration is sometimes preferred to avoid stack overflow and improve performance.

Programming languages that support recursive calls:

- C
- C++
- Java
- Python
- C#
- JavaScript
- Go
- Rust
- Swift
- Kotlin
- Scala
- Haskell
- Lisp
- Scheme
- Pascal